3rd leaf : $\frac{0.3 + 0.3}{(0.1 + (1-0.1)) + (0.1 + (1-0.1))}$ ≈ 1.42

$= \frac{0.6}{0.42}$

Final log(odds) : log(odds) + $\sum_{b=1}^{B} \lambda \cdot \log(\text{odds})_b$

Kernel method ⇒ fit non linear model

It is possible to move a data into higher dimension to fit a data

$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \dots$

$x \in \mathbb{R}^2 \quad \theta \Rightarrow \mathbb{R}^2 \qquad \theta \Rightarrow \mathbb{R}^d$

$p >> d$

$x \in \mathbb{R} \qquad \underbrace{\phi(x)}_{\text{function move data into 2D into 4D}} = \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \end{bmatrix} \Rightarrow \theta \Rightarrow \mathbb{R}^p$

features

before transformation attribute

After transformation feature

$\theta_j := \theta_j - \alpha \sum_{i=1}^{n} (h_\theta(x^{(i)}) - y^{(i)}) \cdot x_j^{(i)}$

$= \theta_j + \alpha \sum_{i=1}^{n} (y^{(i)} - \underbrace{h_\theta x^{(i)}}_{\theta^T \phi(x^{(i)})}) \cdot x_j^{(i)}$

$= \theta_j + \alpha \sum_{i=1}^{n} (y^{(i)} - \underbrace{\theta^T}_{\theta \in \mathbb{R}^p} (\phi x^{(i)})) \phi(x^{(i)})$

$f(x) \}$ → feature map

↳ next, next page

$d = 1000 \quad \theta \in \mathbb{R}^{1000} \quad x \in \mathbb{R}^{1000}$

$$\phi(x) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x^2 \\ x_1 x_2 \\ x_1 x_3 \\ \vdots \\ x_2 x_1 \end{bmatrix}$$ monomial of x of degree ≤ 3

$\approx O(d^3)$

1000 000 000

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 \quad z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \in \mathbb{R}^3 \quad x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad z = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\underbrace{\phi x}_{\text{feature map}} = \begin{bmatrix} x_1 x_2 \\ x_1 x_3 \\ x_1 x_1 \\ x_2 x_1 \\ x_2 x_2 \\ x_2 x_3 \\ x_3 x_1 \\ x_3 x_2 \\ x_3 x_3 \end{bmatrix} \quad \phi z = \begin{bmatrix} z_1 z_1 \\ z_1 z_2 \\ z_1 z_3 \\ z_2 z_1 \\ z_2 z_2 \\ z_2 z_3 \\ z_3 z_1 \\ z_3 z_2 \\ z_3 z_3 \end{bmatrix} \quad \phi(x) = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 4 \\ 6 \\ 3 \\ 6 \\ 9 \end{bmatrix}_{9\times 1} \quad \phi(z) = \begin{bmatrix} 16 \\ 20 \\ 24 \\ 20 \\ 25 \\ 30 \\ 24 \\ 30 \\ 36 \end{bmatrix}_{9\times 1}$$

$$\phi(x) \cdot \phi(z) = 1024$$

Without going higher dimensional space we can get by original dimensional by kernal function

Kernal function

$$k(x, z) = (\langle x, z \rangle)^2$$
$$= (4 + 10 + 18)^2$$
$$= 1024$$

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad z = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \Rightarrow \mathbb{R}^2$$

$$\phi(x) = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2} x_1 x_2 \\ \sqrt{2} x_1 \\ \sqrt{2} x_2 \\ 1 \end{bmatrix}_{6\times 1} \quad \mathbb{R}^6$$

$$k = (x \cdot z + 1)^2$$
$$= (3 + 8 + 1)^2$$
$$= 144$$

$$= \begin{bmatrix} 1 \\ 4 \\ 2.82 \\ 1.41 \\ 2.82 \\ 1 \end{bmatrix} \begin{bmatrix} 9 \\ 16 \\ 16.97 \\ 4.24 \\ 5.65 \\ 1 \end{bmatrix}$$

$$9 + 64 + \cancel{47.85}\ 23.92 + 5.97 + 15.93 + 1$$
$$47.85 = \cancel{+90.75} \quad 143.75$$

Linear Kernal $\Rightarrow k(x, z) = (x \cdot z)$

Polynomial " $\Rightarrow k(x, z) = (x \cdot z + c)^d$

Fisher " $\Rightarrow$

Graph " $\Rightarrow$

Radial Basis (RBF) function $\Rightarrow$

Sinc Kernel $\Rightarrow$

Neural Network, Gaussian process $\Rightarrow$

$$\cancel{\beta_i = \beta_i + \alpha \left[y^{(i)} - \sum_{j=1}^{n} \beta_j \langle \phi(x^j), x^{(i)} \rangle \right.}$$

$$\beta_i = \beta_i + \alpha \left[y^{(i)} - \sum_{j=1}^{n} \beta_j \langle \underbrace{\phi(x^j)(x^i)}_{\text{Inner product of two vectors}} \rangle \right]$$

$$K(x, z) = \langle \phi(x), \phi(z) \rangle$$

$$= 1 + \langle x, z \rangle + \langle x, z \rangle^2 + \langle x, z \rangle^3$$

$$\beta_i = \beta_i + \alpha \left[y^{(i)} - \sum_{j=1}^{n} \beta_j \, K \langle x^{(i)}, x^{(j)} \rangle \right]$$

$$\forall i \in \{1, 2, \ldots, n\}$$

<u>Inference phase</u>

for new unseen $x \in \mathbb{R}^d$

$$h_\theta(x) = \theta^T \phi(x^{(t)})$$

$$\theta = \sum_{i=1}^{n} \beta_i \phi(x^{(i)})$$

$$h_\theta(x) = \theta^T \phi(x) = \sum_{i=1}^{n} \beta_i K(x^{(i)}, x)$$

Train

x_1	x_2	y
1	1	2
3	1	4

Test $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$h_\theta(x) = \theta^T \phi(x)$$

$$h_\theta(x) = \sum_{i=1}^{n} \beta_i K \langle x^{(i)}, x \rangle$$

$\beta = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ assumed Beta

Kernel: Radial Basis function (RBF)

$$K(x, z) = \exp(-\|x - z\|^2) \quad \text{where } \|x - z\|^2 = (x_1 - z_1)^2 + (x_2 - z_2)^2$$

$$h_\theta(x) = \beta_1 \exp(-(1-2)^2 + (1-1)^2) + \beta_2 \exp(-(3-2)^2 + (1-1)^2)$$

$$= \beta_1 \exp^{(-(1^2+2^2))} + \beta_2 \exp^{(-(1)^2+0)}$$

$$= \beta_1 \exp^{(-5)} + \beta_2 \exp^{-1}$$

$$= (1)(0.0067) + (1)\,0.367$$

$$= \cancel{0.374579} = 0.734$$